

LinkBuilder

Outreach automation CRM

A link-building team's whole pipeline, run by one operator: finding donor sites, outreach, AI-drafted replies, payments and placement monitoring. Every AI draft passes versioned contracts before it can be sent.

11,000+

backend tests, full run green

69

contract rules on every AI draft

16,000+

placed links tracked

20+

queue workers in one stack

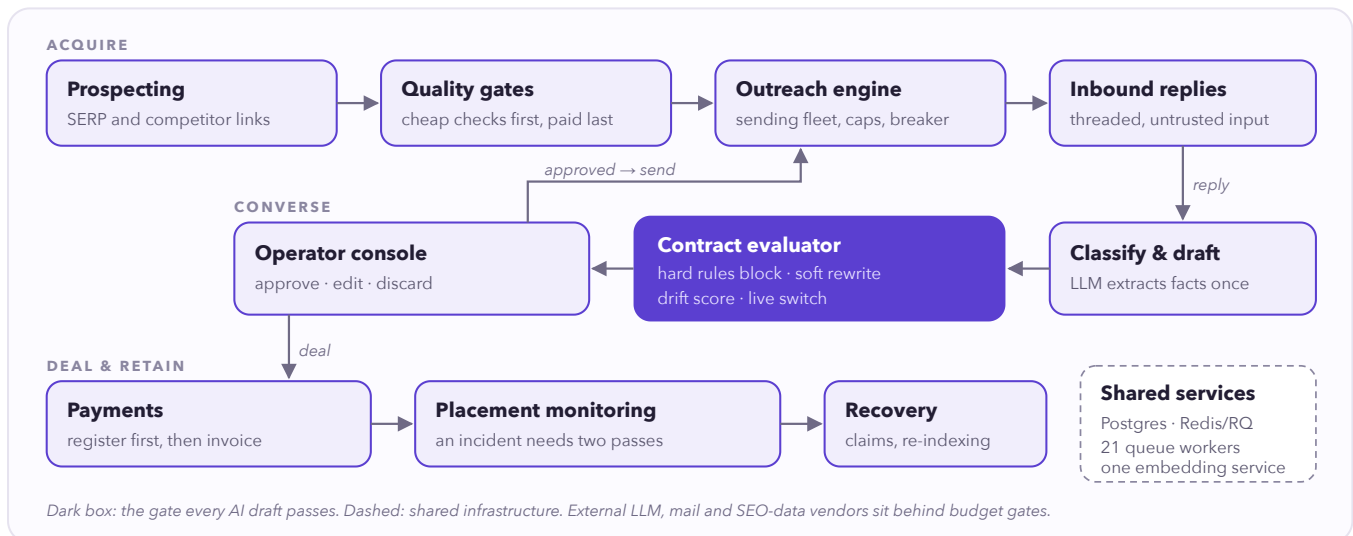
THE PROBLEM

Link building is a long chain of paid, stateful steps, and failures look like successes: a drained data vendor once let a job report "600/600, 0 errors" after about 120 keywords. An LLM writing to strangers about money must obey rules that can be enforced, audited and switched off live.

WHAT IT DOES

- Mines competitors' backlinks and gates every site, the paid check last
- Sends from a fleet of domains with caps and a bounce breaker
- Classifies each reply, drafts an answer, checks it against contracts
- Registers the payment, then keeps verifying the placement

ARCHITECTURE



HOW IT FLOWS

- 1 Acquire** – discover sites, gate them, enrich contacts, launch a campaign.
- 2 Converse** – reply → classify → draft → contract check → a human approves → send.
- 3 Deal & retain** – payment runs register-first; an incident opens only after a confirming pass.
- 4 Control** – drift score and vendor balances raise alerts; a live switch sets full, warn-only or off.

WORTH KNOWING

63 of 69 rules are plain code.

After seven rounds of prompt tuning, the LLM became a fact extractor and deterministic code became the judge.

Safe is not the same as useful.

The safety judge passed 109 of 109 shadow drafts; the operator sent only a quarter unchanged. Auto-send waits for replay data.

"Dead" is checked twice.

159 of 865 "dead" placements were alive – donors' firewalls were blocking the checker, not the links.

DECISIONS THAT SHAPE IT

Correctness in the database, not the scheduler

01

Because one hung send tick once stopped mail for four hours. A send is now claimed atomically and its message ID stored before the provider is called; a stuck send is re-queued only if the provider provably never took it.

Cost: hand-built retries and reapers.

Contracts as versioned specs, not prompt rules

02

Because the rules had drifted between a generator prompt and a validator prompt. A violation now forces a rewrite; if the last attempt still breaks a hard rule, the draft is blocked and escalated.

Cost: 69 invariants and 11 caps to maintain.

Code judges, the LLM extracts

03

Because LLM-as-judge kept its false positives through seven tuning rounds. The model now extracts thread facts once; code rules on them, and tests replay exactly.

Cost: one structured LLM call per thread.

Autonomy earned by measurement

04

Because passing the safety judge is not the same as being worth sending. Auto-send stays off until replay shows 70-80% of first replies sent unchanged.

Cost: operator time on every reply until then.

Register first, then the side effect

05

Because a failed invoice upload once rolled back a registration, and a second request went in for the same money. Registration now commits first, and a test pins the order.

Cost: a visible "registered, no invoice" state to retry.

Fail loud on paid APIs

06

Because a silent partial result costs more than a refused run. A drained vendor now stops the job with a reason, and runs refuse to start without balance.

Cost: runs refused when a balance is low.

HOW IT IS ENGINEERED

Design doc

62 of them, one per feature

→

ADR

with revisit criteria

→

Build

with AI coding agents

→

Gates

16 on commit, 11 ratchets

→

Verification

16 reports and audits

→

Handover

the running system, not the design

11,000+

backend tests, full run green

705

frontend tests

1,098 / 1,953

commits co-written with AI agents, under the gates

84

adversarial prompt-injection cases

Inbound mail is untrusted input. Five layers stand between a webmaster's email and the model: a sanitizer with 61 signatures, untrusted-content wrapping, a fail-closed contract gate, output guards against a redirected payment, and an adversarial suite. Attack payloads are logged only as hashes.

WHERE IT TRANSFERS

The pattern is not about SEO. Any business that lets an LLM write to customers – sales, support, collections, procurement – needs the same layers: the model proposes, versioned contracts check deterministically, a human approves until replay data says otherwise, and an admin can stop everything without a deploy. Scale comes from the boring parts: a stateless API, queue workers that scale independently, correctness held by atomic database claims, and budget caps on every paid call.

STACK

Backend Python, FastAPI, async SQLAlchemy, Alembic, Pydantic, PostgreSQL, Redis + RQ, APScheduler, Playwright · **Frontend** React, TypeScript, Mantine, TanStack Query, Vite · **AI** OpenAI API, a shared MiniLM embedding service · **Infra** Docker Compose

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Built solo for an employer; partner and data details stay private.